As shown in Figure 2, in a private blockchain, we have designated roles like network owner, read-only user, approver and other participants, and the owner has the rights to grant or revoke roles to the participants, as needed. Hence, this kind of blockchain is used for enterprise solutions in domains like retail or for inventory management platforms, to handle a series of processes through a workflow mechanism.

A public blockchain platform is based on a permissionless ledger transaction, where the platform and ledger is open and shared with the public and can be accessed by anyone. Security is ensured in a permissionless ledger by handling distributed ledger transactions. The ledger copy is distributed across the network as multiple copies, and hence tampering with or correcting of ledger entries requires corrections in *all* the ledger copies in the network.

In a permissionless ledger, users can log in anonymously without revealing their identity and the transaction rate can be controlled through a defined owner group. On the other hand, a permissioned ledger has defined roles like the owner, approver, viewer/reader and administrator. These roles define workflow activities, and the control on transaction processing is ensured in such blockchain platforms with these user roles. There may be more than one user for each role so that the speed of the transaction is faster compared to permissionless ledgers.

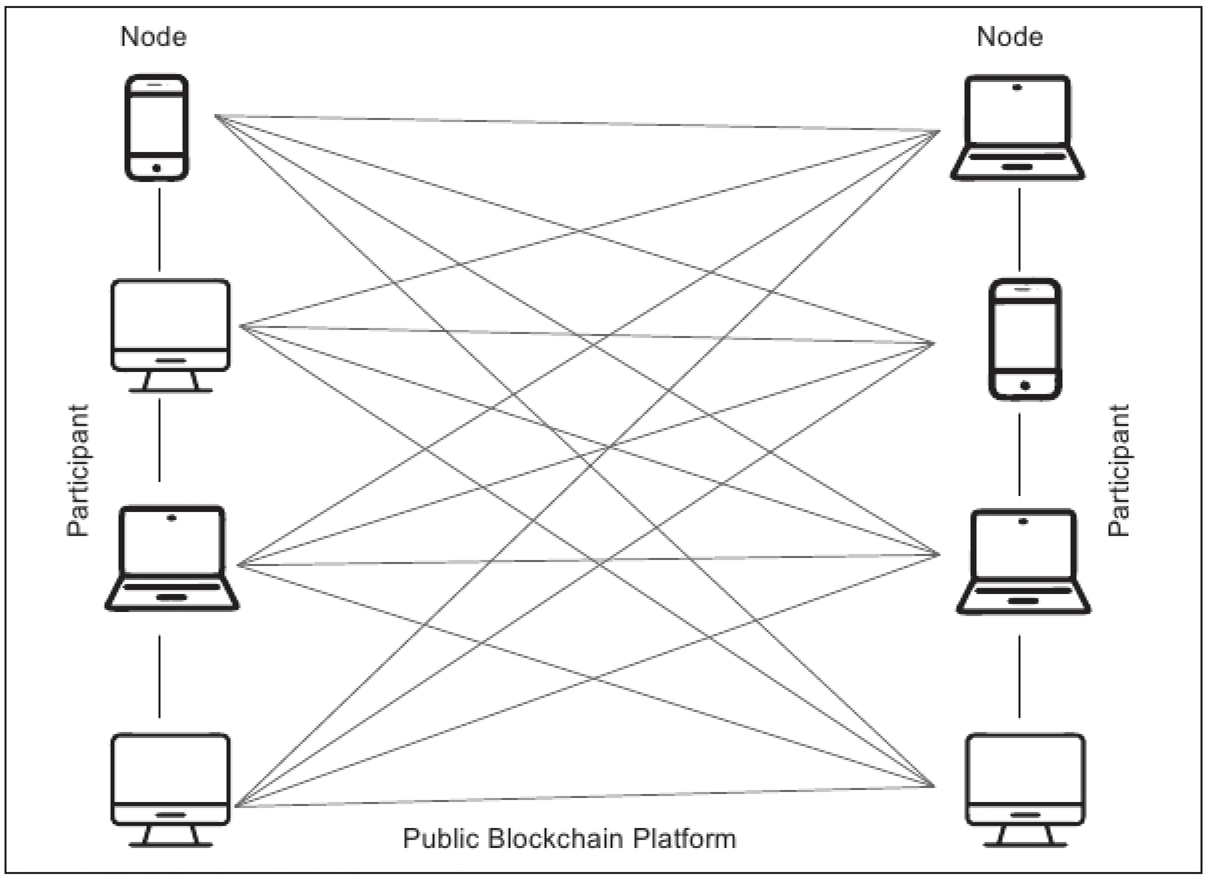

Figure 1: Public blockchain architecture

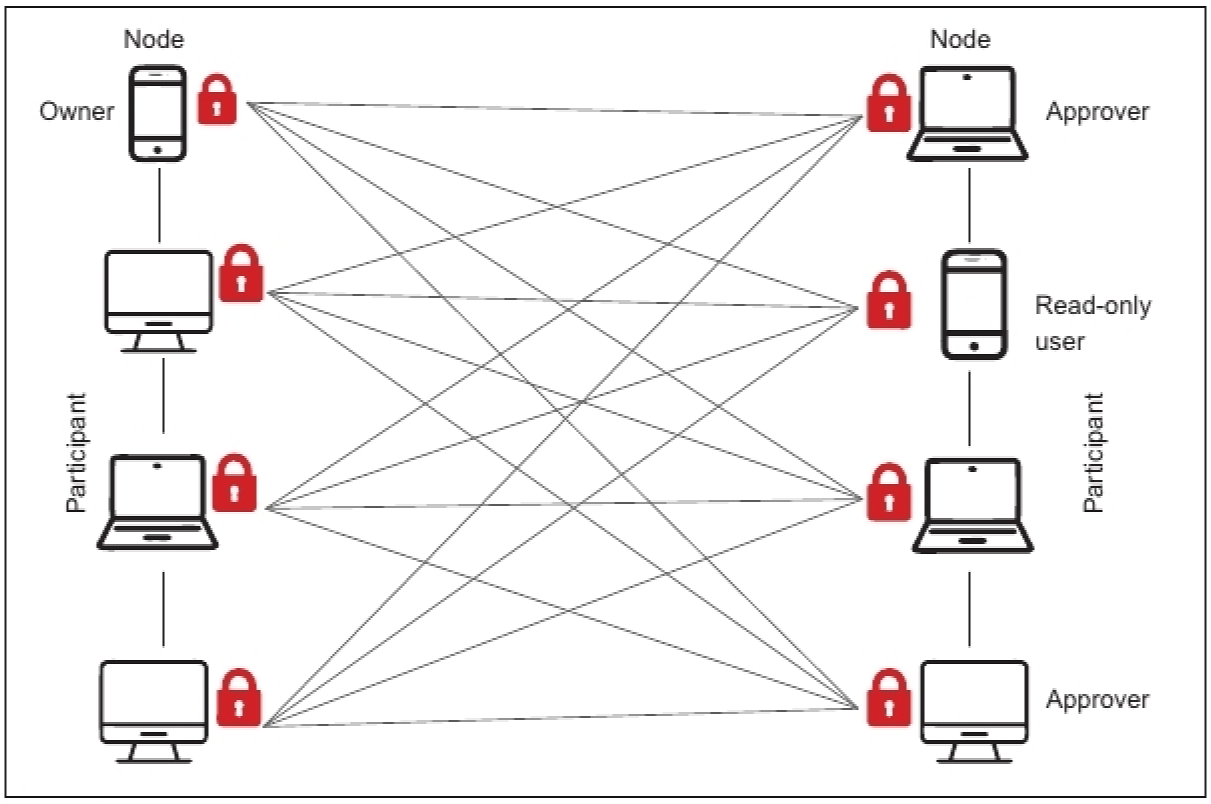

Figure 2: Private blockchain architecture

Security implementation in public and private blockchains

Some popular public blockchain platforms are Ethereum, ZCash, Dash and Litecoin. They have security features incorporated to ensure transactions are safe and the user experience is smooth. They also offer additional benefits like a lower infrastructure cost to run and maintain the decentralised application in the platform.

Some popular private blockchain platforms are R3, Corda and Bitcoin. They have a secured architecture working in a controlled execution platform and, at the same time, ensure reduced transactional costs and data redundancy by reducing semi-manual compliance mechanisms and simplifying document handling activities in transaction processing.

Table 1 helps you understand various factors that affect the blockchain platform in terms of security and architectural needs.

Comparing the performance of public and private blockchains

A public blockchain is slower compared to a private blockchain, which is lightweight and provides higher throughput in transaction processing. Various factors decide the performance of public and private blockchain platforms, as listed below.

Consensus algorithm: This is the step that decides the block to be finalised